ECE26400:
Advanced C Programming

Summer 2026
Course Syllabus

Instructor

Section 001 (CRN: 38194)
Section WNG (CRN: 14473)
Greg Sturm

Prerequisites

CS 15900 (minimum grade of a C-)

Textbook (Optional)

Forouzan and Gilberg, *Computer Science : A Structured Programming Approach Using C (3rd Edition).* Course Technology, 2006, ISBN No. 978-0534491321

Recommended Resources

Lu, Y.-H., & Thiruvathukal, G. K. (2024). *Intermediate C Programming.* (2nd ed.). Taylor & Francis Group. Free Online Access may be found through this link: https://purdue.primo.exlibrisgroup.com/permalink/01PURDUE_PUWL/8j9tnf/alma99170827859901081

Description: Credit Hours: 3.00. Continuation of a first programming course. Topics include files, structures, pointers, and the proper use of dynamic data structures. A basic knowledge of the UNIX operating system and an introductory C programming course; C programming knowledge should include basic syntax, control structures, and file I/O, as well as experience in declaring and using functions.

Course Outcomes

A student who successfully fulfills the course requirements will have demonstrated:

1. the ability to read and write C programs that uses files.
2. the ability to read and write C programs that use structures.
3. the ability to read and write C programs that use dynamic data structures.
4. the ability to read and write C programs that use recursion.

These outcomes form the high-level outline for the material of this course. In more detail, after taking this course you will be able to:

- Explain the organization of a program in memory, including as it executes.

- Explain the principles of data types (including structures) and how they are represented in memory.
- Explain pointers, arrays, strings, and memory management (allocation and deallocation).
- Explain and implement dynamic data structures (linked lists and trees) and basic algorithms over these
- structures.
- Explain the concepts of induction and recursion, and use them to implement recursive functions.
- Explain and implement basic sorting algorithms (both recursive and iterative).
- Explain the concepts of asymptotic analysis and use them to reason about the running time of different algorithms.
- Write programs that read and write from files.

Course Grading

Grades will be assigned as follows:

25% - Midterm exam
35% - Final exam
10% - Lecture exercises
30% - Homeworks

Grade	Cut-off %
A+	97.0
A	93.0
A-	90.0
B+	87.0
B	83.0
B-	80.0
C+	77.0
C	73.0
C-	70.0
D+	67.0
D	63.0
D-	60.0
F	<60.0

Final grades will be truncated, not rounded up. Final grades may be curved up, but will not be curved down.

Homework Assignments

Homework, in the form of programming assignments, will be due most weeks. All homeworks are due on a Saturday night by midnight EST. Your lowest homework score will be dropped.

You can access homework instructions and templates on Brightspace or Github. Programming assignments will be submitted for grading on Gradescope.

Late submissions will be scaled according to lateness, **docking 20% from your score per day late.**

Oral Exam
If I determine there are substantial similarities between the graded work of two students, or determines graded work is substantially similar to work created by an AI tool, I may replace the graded work in question and/or future graded work with an oral exam(s) for the student(s).

Exams
There will be two closed-book, 1-hour exams administered online, proctored at the regularly scheduled class time. You may bring a one-sided, 8.5”x11” handwritten crib sheet to each exam. Please note “handwritten” does not include notes written on a computer and then printed.

For each exam, you must have:
- Access to a reliable internet connection
- Access to a computer for displaying the exam
- Pen/dark pencil and several pieces of paper for writing answers
- Webcam for exam proctoring

Missing an exam/starting an exam late will result in a grade of 0 for that exam. Exam accommodations will only be granted if you demonstrate that it is absolutely necessary (accompanied by verification) and obtain written approval from me in advance of the exam. Justification for exam accommodation includes being in a time zone where the exam would begin at a time between 9:00 PM and 7:00 AM.

Tentative Course Schedule
Below is a rough schedule of when and what will be covered in class:

Week	Topics covered	Assignments Due
1 **06/15**	Introduction Tools Files Programming Layout in Memory	**Homework 1 (6/20)**

Week	Topics covered	Assignments Due
2 **06/22**	Pointers and Arrays Strings Memory Allocation Structs	**Homework 2 (6/27)** **Homework 3 (6/27)** **Homework 4 (6/27)**
3 **06/29**	Bitwise Operations Linked Lists/Stacks and Queues **Programming Practicum #1** **July 4 (No Lecture)**	**Homework 5 (7/4)** **Homework 6 (7/4)**
4 **07/06**	Recursion **Programming Practicum #2** Exam review **Midterm Exam (7/10 at 8:40-9:40 AM EST on Honorlock)**	
5 **07/13**	Sorting Algorithms Depth-First Search Breadth-First Search **Programming Practicum #3**	**Homework 7 (7/18)** **Homework 8 (7/18)**
6 **07/20**	Binary Trees Binary Search Trees Tree Traversal **Programming Practicum #4**	**Homework 9 (7/25)** **Homework 10 (7/25)**
7 **07/27**	Huffman Compression **Programming Practicum #5**	**Homework 11 (8/1)** **Homework 12 (8/1)**
8 **08/03**	Runtime Analysis Conclusion Exam review **Final Exam (8/7 at 8:40-9:40 AM EST on Honorlock)**	